

Manual Change History (3666266) ISM and QSM11 Electronic Control System Troubleshooting and Repair Manual

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Update Reason	

Date	05-Jan-2004
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Update Reason	
Date	05-Jan-2004
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Update Reason	
Date	05-Jan-2004
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Date	06-Jan-2004
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Update Reason	

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Update Reason	

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Update Reason	

Date	09-Jan-2004
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Update Reason	

Date	09-Jan-2004
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Date	12-Jan-2004
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc431 (/qs3/pubsys2/xml/en/procedures/82/82-fc431.html)
Procedure Title	Idle Validation Switch Choice
Update Reason	

Date	12-Jan-2004
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc551 (/qs3/pubsys2/xml/en/procedures/82/82-fc551.html)
Procedure Title	Idle Validation Switch Choice
Update Reason	

Date	16-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666266-t083-tr (/qs3/pubsys2/xml/en/manual/3666266/3666266-t083-tr.html)
Procedure Title	Fault Code Warning Lamps Stay On (No Apparent Reason)
Update Reason	
Date	16-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666266-t091-tr (/qs3/pubsys2/xml/en/manual/3666266/3666266-t091-tr.html)
Procedure Title	Fuel in Coolant
Update Reason	
Date	27-Jan-2004
Section	Back
Procedure Number	3666266-backcover (/qs3/pubsys2/xml/en/manual/3666266/3666266-backcover.html)
Procedure Title	Backcover
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	

Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	
Date	09-Jun-2004
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-208 (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html)
Procedure Title	Deutsch™ HDP20 and HD30 Connector Series
Update Reason	
Date	18-Aug-2004
Section	Section 19 - Electronic Engine Controls
Procedure Number	105-019-063 (/qs3/pubsys2/xml/en/procedures/105/105-019-063.html)
Procedure Title	Internal Actuator Wiring Harness
Update Reason	

Date	15-Dec-2004
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-050 (/qs3/pubsys2/xml/en/procedures/82/82-019-050.html)
Procedure Title	Fuel Shutoff Valve Solenoid
Update Reason	
Date	07-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc254 (/qs3/pubsys2/xml/en/procedures/82/82-fc254.html)
Procedure Title	Fuel Shutoff Solenoid Supply Circuit
Update Reason	
Date	27-Jan-2005
Section	Section F - Familiarization
Procedure Number	82-101-007 (/qs3/pubsys2/xml/en/procedures/82/82-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	
Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc312 (/qs3/pubsys2/xml/en/procedures/82/82-fc312.html)
Procedure Title	Injector Circuit
Update Reason	
Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc313 (/qs3/pubsys2/xml/en/procedures/82/82-fc313.html)
Procedure Title	Injector Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc314 (/qs3/pubsys2/xml/en/procedures/82/82-fc314.html)
Procedure Title	Injector Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc315 (/qs3/pubsys2/xml/en/procedures/82/82-fc315.html)
Procedure Title	Injector Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc321 (/qs3/pubsys2/xml/en/procedures/82/82-fc321.html)
Procedure Title	Injector Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc341 (/qs3/pubsys2/xml/en/procedures/82/82-fc341.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc343 (/qs3/pubsys2/xml/en/procedures/82/82-fc343.html)
Procedure Title	Electronic Control Module (ECM)
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc352 (/qs3/pubsys2/xml/en/procedures/82/82-fc352.html)
Procedure Title	Sensor Voltage Supply
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc359 (/qs3/pubsys2/xml/en/procedures/82/82-fc359.html)
Procedure Title	ICON™ Engine Auto Start Failure
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc386 (/qs3/pubsys2/xml/en/procedures/82/82-fc386.html)
Procedure Title	Sensor Voltage Supply
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc388 (/qs3/pubsys2/xml/en/procedures/82/82-fc388.html)
Procedure Title	Engine Brake Supply Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc392 (/qs3/pubsys2/xml/en/procedures/82/82-fc392.html)
Procedure Title	Engine Brake Supply Circuit
Update Reason	

Date	27-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc419 (/qs3/pubsys2/xml/en/procedures/82/82-fc419.html)
Procedure Title	Intake Manifold Pressure Sensor
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc217 (/qs3/pubsys2/xml/en/procedures/82/82-fc217.html)
Procedure Title	Wet Tank Pressure Sensor Circuit
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc221 (/qs3/pubsys2/xml/en/procedures/82/82-fc221.html)
Procedure Title	Ambient Air Pressure Sensor Circuit
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc222 (/qs3/pubsys2/xml/en/procedures/82/82-fc222.html)
Procedure Title	Ambient Air Pressure Sensor Circuit
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc223 (/qs3/pubsys2/xml/en/procedures/82/82-fc223.html)
Procedure Title	Burn Valve Solenoid Fault
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc227 (/qs3/pubsys2/xml/en/procedures/82/82-fc227.html)
Procedure Title	Sensor Supply Voltage
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc241 (/qs3/pubsys2/xml/en/procedures/82/82-fc241.html)
Procedure Title	Vehicle Speed Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc242 (/qs3/pubsys2/xml/en/procedures/82/82-fc242.html)
Procedure Title	Vehicle Speed Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc245 (/qs3/pubsys2/xml/en/procedures/82/82-fc245.html)
Procedure Title	Engine Fan Clutch Supply Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc249 (/qs3/pubsys2/xml/en/procedures/82/82-fc249.html)
Procedure Title	Ambient Air Temperature Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc255 (/qs3/pubsys2/xml/en/procedures/82/82-fc255.html)
Procedure Title	Fuel Shutoff Solenoid Supply Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc256 (/qs3/pubsys2/xml/en/procedures/82/82-fc256.html)
Procedure Title	Ambient Air Temperature Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc285 (/qs3/pubsys2/xml/en/procedures/82/82-fc285.html)
Procedure Title	SAE J1939 Data Link Multiplexing
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc293 (/qs3/pubsys2/xml/en/procedures/82/82-fc293.html)
Procedure Title	OEM Temperature Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc294 (/qs3/pubsys2/xml/en/procedures/82/82-fc294.html)
Procedure Title	OEM Temperature Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc295 (/qs3/pubsys2/xml/en/procedures/82/82-fc295.html)
Procedure Title	Ambient Air Pressure Sensor
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc311 (/qs3/pubsys2/xml/en/procedures/82/82-fc311.html)
Procedure Title	Injector Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc319 (/qs3/pubsys2/xml/en/procedures/82/82-fc319.html)
Procedure Title	Real Time Clock Power Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc422 (/qs3/pubsys2/xml/en/procedures/82/82-fc422.html)
Procedure Title	Coolant Level Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc428 (/qs3/pubsys2/xml/en/procedures/82/82-fc428.html)
Procedure Title	Water In Fuel (WIF) Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc429 (/qs3/pubsys2/xml/en/procedures/82/82-fc429.html)
Procedure Title	Water In Fuel (WIF) Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc435 (/qs3/pubsys2/xml/en/procedures/82/82-fc435.html)
Procedure Title	Oil Pressure Sensor
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc441 (/qs3/pubsys2/xml/en/procedures/82/82-fc441.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc465 (/qs3/pubsys2/xml/en/procedures/82/82-fc465.html)
Procedure Title	Wastegate Actuator No. 1 Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc466 (/qs3/pubsys2/xml/en/procedures/82/82-fc466.html)
Procedure Title	Wastegate Actuator No. 1 Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc469 (/qs3/pubsys2/xml/en/procedures/82/82-fc469.html)
Procedure Title	ICON™ Cab Thermostat Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc472 (/qs3/pubsys2/xml/en/procedures/82/82-fc472.html)
Procedure Title	Crankcase Oil Level Sensor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc474 (/qs3/pubsys2/xml/en/procedures/82/82-fc474.html)
Procedure Title	Starter Solenoid Lockout Relay Driver Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc475 (/qs3/pubsys2/xml/en/procedures/82/82-fc475.html)
Procedure Title	Electronic Air Compressor Governor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc476 (/qs3/pubsys2/xml/en/procedures/82/82-fc476.html)
Procedure Title	Electronic Air Compressor Governor Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc489 (/qs3/pubsys2/xml/en/procedures/82/82-fc489.html)
Procedure Title	Auxilliary Speed Input Error
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc491 (/qs3/pubsys2/xml/en/procedures/82/82-fc491.html)
Procedure Title	Wastegate Actuator No. 2 Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc492 (/qs3/pubsys2/xml/en/procedures/82/82-fc492.html)
Procedure Title	Wastegate Actuator No. 2 Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc527 (/qs3/pubsys2/xml/en/procedures/82/82-fc527.html)
Procedure Title	Switched Output A Error
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc528 (/qs3/pubsys2/xml/en/procedures/82/82-fc528.html)
Procedure Title	Switched Output B Error
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc529 (/qs3/pubsys2/xml/en/procedures/82/82-fc529.html)
Procedure Title	Switched Output C Error
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc536 (/qs3/pubsys2/xml/en/procedures/82/82-fc536.html)
Procedure Title	Autoshift Low Gear Actuator (Lockout Solenoid)
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc537 (/qs3/pubsys2/xml/en/procedures/82/82-fc537.html)
Procedure Title	Autoshift High Gear Actuator (Lockout Solenoid)
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc538 (/qs3/pubsys2/xml/en/procedures/82/82-fc538.html)
Procedure Title	Autoshift Neutral Actuator
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc541 (/qs3/pubsys2/xml/en/procedures/82/82-fc541.html)
Procedure Title	ICON™ Starter Relay Input Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc544 (/qs3/pubsys2/xml/en/procedures/82/82-fc544.html)
Procedure Title	Top 2 Transmission Circuit - Mechanical System Failure
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc551sss (/qs3/pubsys2/xml/en/procedures/82/82-fc551sss.html)
Procedure Title	Idle Validation Switch Circuit - Solid State Switch and Sensor Type
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc581 (/qs3/pubsys2/xml/en/procedures/82/82-fc581.html)
Procedure Title	Fuel Inlet Restriction Sensor Circuit
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc582 (/qs3/pubsys2/xml/en/procedures/82/82-fc582.html)
Procedure Title	Fuel Inlet Restriction Sensor Circuit
Update Reason	
Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc588 (/qs3/pubsys2/xml/en/procedures/82/82-fc588.html)
Procedure Title	ICON™ Starter Relay Input Circuit
Update Reason	

Date	28-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc589 (/qs3/pubsys2/xml/en/procedures/82/82-fc589.html)
Procedure Title	Engine Start Alarm Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-017 (/qs3/pubsys2/xml/en/procedures/82/82-019-017.html)
Procedure Title	Engine Coolant Level Sensor
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-019 (/qs3/pubsys2/xml/en/procedures/82/82-019-019.html)
Procedure Title	Engine Coolant Temperature Sensor
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-021 (/qs3/pubsys2/xml/en/procedures/82/82-019-021.html)
Procedure Title	Cruise Control or PTO ON/OFF Switch
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-031 (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html)
Procedure Title	Electronic Control Module (ECM)
Update Reason	

Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-039 (/qs3/pubsys2/xml/en/procedures/82/82-019-039.html)
Procedure Title	Engine Position Sensor (EPS) Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-045 (/qs3/pubsys2/xml/en/procedures/82/82-019-045.html)
Procedure Title	Fan Clutch Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-049 (/qs3/pubsys2/xml/en/procedures/82/82-019-049.html)
Procedure Title	Fuel Shutoff Valve Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-058 (/qs3/pubsys2/xml/en/procedures/82/82-019-058.html)
Procedure Title	Injector Solenoid Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-108 (/qs3/pubsys2/xml/en/procedures/82/82-019-108.html)
Procedure Title	Intermediate Speed Control Validation Switch
Update Reason	

Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-161 (/qs3/pubsys2/xml/en/procedures/82/82-019-161.html)
Procedure Title	Data Link Circuit, SAE J1939
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-168 (/qs3/pubsys2/xml/en/procedures/82/82-019-168.html)
Procedure Title	Maintenance Lamp Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-169 (/qs3/pubsys2/xml/en/procedures/82/82-019-169.html)
Procedure Title	Electronic Air Compressor Governor
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-303 (/qs3/pubsys2/xml/en/procedures/82/82-019-303.html)
Procedure Title	Starter Relay Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-307 (/qs3/pubsys2/xml/en/procedures/82/82-019-307.html)
Procedure Title	Interlock Input Circuit
Update Reason	

Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-308 (/qs3/pubsys2/xml/en/procedures/82/82-019-308.html)
Procedure Title	Ignition Relay Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-310 (/qs3/pubsys2/xml/en/procedures/82/82-019-310.html)
Procedure Title	Engine Start Alarm Circuit
Update Reason	
Date	28-Jan-2005
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-313 (/qs3/pubsys2/xml/en/procedures/82/82-019-313.html)
Procedure Title	Shaft Speed Sensor Circuit
Update Reason	
Date	31-Jan-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc235 (/qs3/pubsys2/xml/en/procedures/82/82-fc235.html)
Procedure Title	Engine Coolant Level - Engine Protection
Update Reason	
Date	01-Feb-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc147 (/qs3/pubsys2/xml/en/procedures/82/82-fc147.html)
Procedure Title	Frequency Throttle Control
Update Reason	

Date	01-Feb-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc148 (/qs3/pubsys2/xml/en/procedures/82/82-fc148.html)
Procedure Title	Frequency Throttle Control
Update Reason	
Date	01-Feb-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc153 (/qs3/pubsys2/xml/en/procedures/82/82-fc153.html)
Procedure Title	Intake Manifold Air Temperature Sensor Circuit
Update Reason	
Date	01-Feb-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc216 (/qs3/pubsys2/xml/en/procedures/82/82-fc216.html)
Procedure Title	Wet Tank Pressure Sensor Circuit
Update Reason	
Date	09-Jan-2006
Section	Section V - Specifications
Procedure Number	4021559 (/qs3/pubsys2/xml/en/procedures//4021559.html)
Procedure Title	Specification
Update Reason	
Date	06-Feb-2006
Section	Section 19 - Electronic Engine Controls
Procedure Number	105-019-428 (/qs3/pubsys2/xml/en/procedures/105/105-019-428.html)
Procedure Title	Engine Datalinks
Update Reason	

Date	12-May-2006
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-127 (/qs3/pubsys2/xml/en/procedures/82/82-019-127.html)
Procedure Title	Water In Fuel Sensor
Update Reason	
Date	12-May-2006
Section	Section 19 - Electronic Engine Controls
Procedure Number	82-019-135 (/qs3/pubsys2/xml/en/procedures/82/82-019-135.html)
Procedure Title	Ambient Air Temperature Sensor Circuit
Update Reason	
Date	22-May-2006
Section	Front
Procedure Number	tr-frwd (/qs3/pubsys2/xml/en/manual/preface/tr-frwd.html)
Procedure Title	Foreword
Update Reason	
Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	20-Nov-2006
Section	Section V - Specifications
Procedure Number	3666269 (/qs3/pubsys2/xml/en/procedures//3666269.html)
Procedure Title	Specification
Update Reason	

Date	20-Nov-2006
Section	Section V - Specifications
Procedure Number	3666413 (/qs3/pubsys2/xml/en/procedures//3666413.html)
Procedure Title	Specification
Update Reason	
Date	18-Dec-2006
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc443 (/qs3/pubsys2/xml/en/procedures/82/82-fc443.html)
Procedure Title	Accelerator Pedal Voltage Supply
Update Reason	
Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	
Date	24-Jul-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc187 (/qs3/pubsys2/xml/en/procedures/82/82-fc187.html)
Procedure Title	Sensor Supply Voltage
Update Reason	
Date	24-Jul-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc237 (/qs3/pubsys2/xml/en/procedures/82/82-fc237.html)
Procedure Title	Multiple Unit Synchronization (soft coupled Marine)
Update Reason	

Date	18-Oct-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc297 (/qs3/pubsys2/xml/en/procedures/82/82-fc297.html)
Procedure Title	OEM Pressure Sensor Circuit
Update Reason	
Date	18-Oct-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc298 (/qs3/pubsys2/xml/en/procedures/82/82-fc298.html)
Procedure Title	OEM Pressure Sensor Circuit
Update Reason	
Date	05-Nov-2007
Section	Section V - Specifications
Procedure Number	3666468 (/qs3/pubsys2/xml/en/procedures//3666468.html)
Procedure Title	Specification
Update Reason	
Date	29-Nov-2007
Section	Section V - Specifications
Procedure Number	4021694 (/qs3/pubsys2/xml/en/procedures//4021694.html)
Procedure Title	Specification
Update Reason	
Date	30-Nov-2007
Section	Section V - Specifications
Procedure Number	4081953 (/qs3/pubsys2/xml/en/procedures//4081953.html)
Procedure Title	Specification
Update Reason	

Date	07-Dec-2007
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	
Date	03-Jan-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc433 (/qs3/pubsys2/xml/en/procedures/82/82-fc433.html)
Procedure Title	Intake Manifold Pressure Sensor Circuit
Update Reason	
Date	28-Jan-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)
Procedure Title	Resistance Measurement Using a Multimeter
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-008 (/qs3/pubsys2/xml/en/procedures/99/99-019-008.html)
Procedure Title	Battery Ground Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-022 (/qs3/pubsys2/xml/en/procedures/99/99-019-022.html)
Procedure Title	Cruise Control or PTO ON/OFF Switch Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-024 (/qs3/pubsys2/xml/en/procedures/99/99-019-024.html)
Procedure Title	Cruise Control or PTO Set/Resume Select Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-028 (/qs3/pubsys2/xml/en/procedures/99/99-019-028.html)
Procedure Title	Diagnostic Test Mode Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-034 (/qs3/pubsys2/xml/en/procedures/99/99-019-034.html)
Procedure Title	Engine Brake ON/OFF Switch
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-035 (/qs3/pubsys2/xml/en/procedures/99/99-019-035.html)
Procedure Title	Engine Brake ON/OFF Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-047 (/qs3/pubsys2/xml/en/procedures/99/99-019-047.html)
Procedure Title	Fault Lamp Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-053 (/qs3/pubsys2/xml/en/procedures/99/99-019-053.html)
Procedure Title	Idle Adjust Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-071 (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)
Procedure Title	OEM Wiring Harness
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-078 (/qs3/pubsys2/xml/en/procedures/99/99-019-078.html)
Procedure Title	Programmable Features and Parameters Not Correct
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-079 (/qs3/pubsys2/xml/en/procedures/99/99-019-079.html)
Procedure Title	Remote PTO Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-089 (/qs3/pubsys2/xml/en/procedures/99/99-019-089.html)
Procedure Title	Brake Pedal Position Switch Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-090 (/qs3/pubsys2/xml/en/procedures/99/99-019-090.html)
Procedure Title	Vehicle Speed Sensor, Digital Input
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-091 (/qs3/pubsys2/xml/en/procedures/99/99-019-091.html)
Procedure Title	Vehicle Speed Sensor, Magnetic Pick Up
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-093 (/qs3/pubsys2/xml/en/procedures/99/99-019-093.html)
Procedure Title	Vehicle Speed Sensor Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-166 (/qs3/pubsys2/xml/en/procedures/99/99-019-166.html)
Procedure Title	Data Link Circuit, SAE J1587
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-261 (/qs3/pubsys2/xml/en/procedures/99/99-019-261.html)
Procedure Title	Air Conditioning Pressure Switch
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-262 (/qs3/pubsys2/xml/en/procedures/99/99-019-262.html)
Procedure Title	Air Conditioning Pressure Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-381 (/qs3/pubsys2/xml/en/procedures/99/99-019-381.html)
Procedure Title	Fan Control Switch Circuit
Update Reason	
Date	11-Jun-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-010 (/qs3/pubsys2/xml/en/procedures/99/99-019-010.html)
Procedure Title	Clutch Pedal Position Switch Circuit
Update Reason	
Date	11-Jun-2008
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-165 (/qs3/pubsys2/xml/en/procedures/99/99-019-165.html)
Procedure Title	Data Link Circuit, SAE J1939
Update Reason	
Date	08-Sep-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc323 (/qs3/pubsys2/xml/en/procedures/82/82-fc323.html)
Procedure Title	Injector Circuit
Update Reason	

Date	08-Sep-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc324 (/qs3/pubsys2/xml/en/procedures/82/82-fc324.html)
Procedure Title	Injector Circuit
Update Reason	
Date	08-Sep-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc325 (/qs3/pubsys2/xml/en/procedures/82/82-fc325.html)
Procedure Title	Injector Circuit
Update Reason	
Date	08-Sep-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc331 (/qs3/pubsys2/xml/en/procedures/82/82-fc331.html)
Procedure Title	Injector Circuit
Update Reason	
Date	08-Sep-2008
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc332 (/qs3/pubsys2/xml/en/procedures/82/82-fc332.html)
Procedure Title	Injector Circuit
Update Reason	
Date	07-Jan-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc951 (/qs3/pubsys2/xml/en/procedures/82/82-fc951.html)
Procedure Title	Cylinder Power Imbalance Between Cylinders - Data Erratic, Intermittent, or Incorrect
Update Reason	

Date	03-Mar-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666266-t046-tr (/qs3/pubsys2/xml/en/manual/3666266/3666266-t046-tr.html)
Procedure Title	Engine Fan Does Not Operate, Operates Erratically, or Operates Continuously
Update Reason	
Date	10-Mar-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666266-t092-tr (/qs3/pubsys2/xml/en/manual/3666266/3666266-t092-tr.html)
Procedure Title	Fuel in Lubricating Oil
Update Reason	
Date	19-Mar-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-214 (/qs3/pubsys2/xml/en/procedures/99/99-019-214.html)
Procedure Title	Bosch™ Actuator and Sensor Connector Series
Update Reason	
Date	19-Mar-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-215 (/qs3/pubsys2/xml/en/procedures/99/99-019-215.html)
Procedure Title	Bosch™ ECM Injector Driver Connector Series
Update Reason	
Date	23-Apr-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-216 (/qs3/pubsys2/xml/en/procedures/99/99-019-216.html)
Procedure Title	Bosch™ ECM Actuator and Sensor Connector Series
Update Reason	

Date	23-Apr-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-217 (/qs3/pubsys2/xml/en/procedures/99/99-019-217.html)
Procedure Title	Bosch™ ECM OEM Connector Series
Update Reason	
Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	
Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	04-Aug-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc322 (/qs3/pubsys2/xml/en/procedures/82/82-fc322.html)
Procedure Title	Injector Circuit
Update Reason	
Date	20-Aug-2009
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	105-t02-1001 (/qs3/pubsys2/xml/en/procedures/105/105-t02-1001.html)
Procedure Title	Engine Performance Troubleshooting Tree
Update Reason	

Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	
Date	16-Sep-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-064 (/qs3/pubsys2/xml/en/procedures/99/99-019-064.html)
Procedure Title	Key Switch Battery Supply Circuit
Update Reason	
Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	20-Oct-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	105-019-427 (/qs3/pubsys2/xml/en/procedures/105/105-019-427.html)
Procedure Title	ECM ROM Boot
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)
Procedure Title	Multimeter Usage
Update Reason	

Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls
Procedure Number	99-019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
Procedure Title	Inactive or Intermittent Fault Code
Update Reason	
Date	11-Dec-2009
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc434 (/qs3/pubsys2/xml/en/procedures/82/82-fc434.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	

Date	18-Feb-2010
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	
Date	19-Mar-2010
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc3714 (/qs3/pubsys2/xml/en/procedures/82/82-fc3714.html)
Procedure Title	Engine Protection Torque Derate - Condition Exists
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	25-May-2010
Section	Section i - Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	
Date	16-Jun-2010
Section	Section 19 - Electronic Engine Controls
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	

Date	23-Jun-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666266-t016-tr (/qs3/pubsys2/xml/en/manual/3666266/3666266-t016-tr.html)
Procedure Title	Communication Error - Electronic Service Tool or Control Device
Update Reason	
Date	17-Aug-2010
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-474 (/qs3/pubsys2/xml/en/procedures/99/99-019-474.html)
Procedure Title	Bosch Aftertreatment DEF Controller Connector
Update Reason	Added procedure to outline.
Date	7-Oct-10
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc143 (/qs3/pubsys2/xml/en/procedures/82/82-fc143.html)
Procedure Title	Engine Oil Rifle Pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level
Update Reason	New fault code released for this ISM manual.
Date	7-Oct-10
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-143 (/qs3/pubsys2/xml/en/procedures/82/82-t05-143.html)
Procedure Title	Engine Oil Rifle Pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level
Update Reason	New fault code released for this ISM manual.

Date	21-Jan-2011
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Updated General Information, Initial Check, Setup, and Resistance Check sections.

Date	15-Aug-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-491 (/qs3/pubsys2/xml/en/procedures/82/82-t05-491.html)
Procedure Title	Turbocharger 2 Wastegate Control Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Added troubleshooting step to check the wastegate actuator.

Date	23-Aug-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc381 (/qs3/pubsys2/xml/en/procedures/82/82-fc381.html)
Procedure Title	Intake Air Heater Number 1 Cercuit Error
Update Reason	Added fault code to outline.

Date	23-Aug-2011
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-381 (/qs3/pubsys2/xml/en/procedures/82/82-t05-381.html)
Procedure Title	Intake Air Heater Number 1 Circuit Error
Update Reason	Added fault code to outline.

Date	10-Feb-2012
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	Updated to include information concerning troubleshooting of ECM calibration issues once new calibration is installed.

Date	12-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-322 (/qs3/pubsys2/xml/en/procedures/82/82-t05-322.html)
Procedure Title	Injector Solenoid Driver Cylinder 1 Circuit - Current Below Normal, or Open Circuit
Update Reason	Changed Guided Step 1A action from "Check for inactive fault codes." to "Check for active fault codes.".

Date	12-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-323 (/qs3/pubsys2/xml/en/procedures/82/82-t05-323.html)
Procedure Title	Injector Solenoid Driver Cylinder 5 Circuit - Current Below Normal, or Open Circuit
Update Reason	Changed Guided Step 1A action from "Check for inactive fault codes." to "Check for active fault codes.".

Date	12-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-324 (/qs3/pubsys2/xml/en/procedures/82/82-t05-324.html)
Procedure Title	Injector Solenoid Driver Cylinder 3 Circuit - Current Below Normal, or Open Circuit
Update Reason	Changed Guided Step 1A action from "Check for inactive fault codes." to "Check for active fault codes.".

Date	12-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-325 (/qs3/pubsys2/xml/en/procedures/82/82-t05-325.html)
Procedure Title	Injector Solenoid Driver Cylinder 6 Circuit - Current Below Normal, or Open Circuit
Update Reason	Changed Guided Step 1A action from "Check for inactive fault codes." to "Check for active fault codes.".

Date	12-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-331 (/qs3/pubsys2/xml/en/procedures/82/82-t05-331.html)
Procedure Title	Injector Solenoid Driver Cylinder 2 Circuit - Current Below Normal, or Open Circuit
Update Reason	Changed Guided Step 1A action from "Check for inactive fault codes." to "Check for active fault codes.".

Date	12-Jun-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-332 (/qs3/pubsys2/xml/en/procedures/82/82-t05-332.html)
Procedure Title	Injector Solenoid Driver Cylinder 4 Circuit - Current Below Normal, or Open Circuit
Update Reason	Changed Guided Step 1A action from "Check for inactive fault codes." to "Check for active fault codes.".

Date	01-Nov-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc465 (/qs3/pubsys2/xml/en/procedures/82/82-fc465.html)
Procedure Title	Wastegate Actuator Number 1 Circuit
Update Reason	Shop Talk: add, Possible causes of this fault code.

Date	01-Nov-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc491 (/qs3/pubsys2/xml/en/procedures/82/82-fc491.html)
Procedure Title	Wastegate Actuator Number 2 Circuit
Update Reason	Shop Talk: add, Possible causes of this fault code.

Date	01-Nov-2012
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-465 (/qs3/pubsys2/xml/en/procedures/82/82-t05-465.html)
Procedure Title	Turbocharger 1 Wastegate Control Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	Added fault code to manual.

Date	03-Jul-2013
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	3666266-t087-tr (/qs3/pubsys2/xml/en/manual/3666266/3666266-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	Removed Cause: VE/VMS® analysis uses incorrect data

Date	20-Sep-2013
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	82-019-107 (/qs3/pubsys2/xml/en/procedures/82/82-019-107.html)
Procedure Title	Intermediate Speed Control Switch
Update Reason	Replaced this procedure with a common procedure.

Date	20-Sep-2013
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	00-019-107 (/qs3/pubsys2/xml/en/procedures/00/00-019-107.html)
Procedure Title	Intermediate Speed Control Switch
Update Reason	Added to outline.

Date	16-Jan-2014
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-343 (/qs3/pubsys2/xml/en/procedures/82/82-t05-343.html)
Procedure Title	Engine Control Module Warning Internal Hardware Failure - Bad Intelligent Device or Component
Update Reason	Corrected the fault code title.

Date	18-May-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc497 (/qs3/pubsys2/xml/en/procedures/82/82-fc497.html)
Procedure Title	Multiple Unit Synchronization Switch - Data Erratic, Intermittent, or Incorrect
Update Reason	Added fault code to manual.
Date	20-May-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc322 (/qs3/pubsys2/xml/en/procedures/82/82-fc322.html)
Procedure Title	Injector Solenoid Driver Cylinder 1 Circuit - Current Below Normal or Open Circuit
Update Reason	Changed Circuit Art to shown 10 and 9.
Date	20-May-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-322 (/qs3/pubsys2/xml/en/procedures/82/82-t05-322.html)
Procedure Title	Injector Solenoid Driver Cylinder 1 Circuit - Current Below Normal or Open Circuit
Update Reason	Step 2B, changed pins to 9 and 10 at the ECM connector.
Date	04-Aug-2015
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-141 (/qs3/pubsys2/xml/en/procedures/82/82-t05-141.html)
Procedure Title	Engine Oil Pressure/Temperature Sensor Circuit
Update Reason	Updated fault code to replace malfunctioning sensor instead engine control module (ECM).

Date	05-Dec-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	Modified Steps 2, 6, 9 and 11 and added information for new INLINE™ 7 data link.
Date	13-Dec-2016
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Removed or corrected information for INLINE™ 4, 5, and 6. Added new information for INLINE™ 7 data link.
Date	19-Jun-2017
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc415 (/qs3/pubsys2/xml/en/procedures/82/82-fc415.html)
Procedure Title	Engine Oil Rifle Pressure - Data Valid But Below Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.
Date	18-Oct-2018
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc155 (/qs3/pubsys2/xml/en/procedures/82/82-fc155.html)
Procedure Title	Intake Manifold 1 Temperature - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.

Date	22-Jan-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-422 (/qs3/pubsys2/xml/en/procedures/82/82-t05-422.html)
Procedure Title	Coolant Level - Data Erratic, Intermittent, or Incorrect
Update Reason	Clarified the different when a shorting plug is used, and general clean-up.

Date	31-May-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-254 (/qs3/pubsys2/xml/en/procedures/82/82-t05-254.html)
Procedure Title	Engine Fuel Shutoff Valve Driver Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	Revised fault code to standard format. Changed Step 1C-1 'No' step from 2A to 1D.

Date	21-Jun-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-fc214 (/qs3/pubsys2/xml/en/procedures/82/82-fc214.html)
Procedure Title	Engine Oil Temperature - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.

Date	01-Jul-2019
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	82-t05-214 (/qs3/pubsys2/xml/en/procedures/82/82-t05-214.html)
Procedure Title	Engine Oil Temperature - Data Valid But Above Normal Operating Range - Most Severe Level
Update Reason	Added fault code to manual.

Date	21-Oct-2019
Section	Section 19 - Electronic Controls - Group 19
Procedure Number	105-019-032 (/qs3/pubsys2/xml/en/procedures/105/105-019-032.html)
Procedure Title	Engine Control Module Calibration Code
Update Reason	Added a statement to have the technician "verify vehicle odometer is equal to the value recorded before ECM removal."
Date	24-Sep-2024
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	82-019-031 (/qs3/pubsys2/xml/en/procedures/82/82-019-031.html)
Procedure Title	Engine Control Module
Update Reason	No changes made to procedure, needed to update procedure title to align to proper assigned title.

Last Modified: 10-Dec-2024
